

Agents and Tools

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Evidence and Sources

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for State machines for bounded agents. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for State machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid transition.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

6. Solve the given case

An agent may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An agent plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

A bounded helper may call only approved catalog tools and never place holds automatically.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid transition.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Agents and Tools

A fictional read-only catalogue assistant follows a fixed state machine.

Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge.

Trace A: ready→searching→checking→done. Trace B: ready→searching→done.

Agents and Tools

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare transitions

A state is a named stage; only listed edges are allowed.



2. Trace A

Every adjacent edge is listed: valid.



3. Trace B

searching -> done is absent: reject at this first invalid transition.



4. Terminal check

After done, another search requires a separately defined new run.

CHANGE ONE THING

Add a review state between checking and done, without updating Trace A.

Explain the changed result using the matching lesson.

Can the assistant accept Trace B because its final state is done?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Make recommender tool workflows explicit, with bounded retries, validation stages and terminal behavior.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Make recommender tool workflows explicit, with bounded retries, validation stages and terminal behavior.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional read-only catalogue assistant follows a fixed state machine.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? After done, another search requires a separately defined new run.

1. Trace A
2. Trace B
3. Terminal check
4. Declare transitions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Terminal check After done, another search requires a separately defined new run. Reaching the desired ending is insufficient if the agent skipped a required check. Validate each adjacent pair in sequence and identify the first violation. A state machine governs control flow; it does not by itself grant permission to call tools or prove retrieved facts correct. Those are separate checks inside the relevant stages.

Practice 2: explain the deciding evidence

Changed-case practice: Add a review state between checking and done, without updating Trace A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Trace A becomes invalid at checking→done under the revised table.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Trace A becomes invalid at checking→done under the revised table. A required new transition changes what constitutes a complete run.

Repair a missing building block

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
